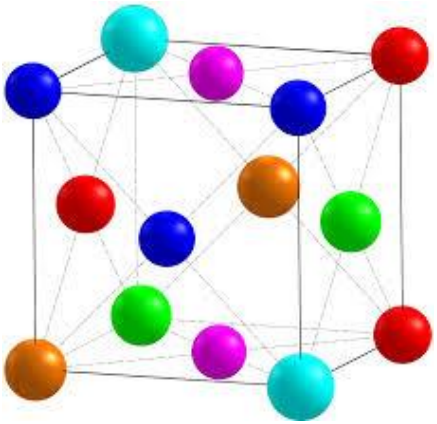
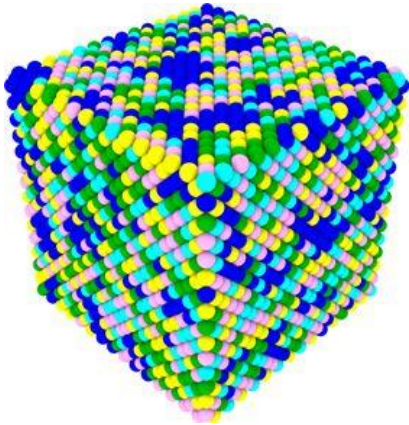
MATERIAL SCIENCE AND ENGINEERING

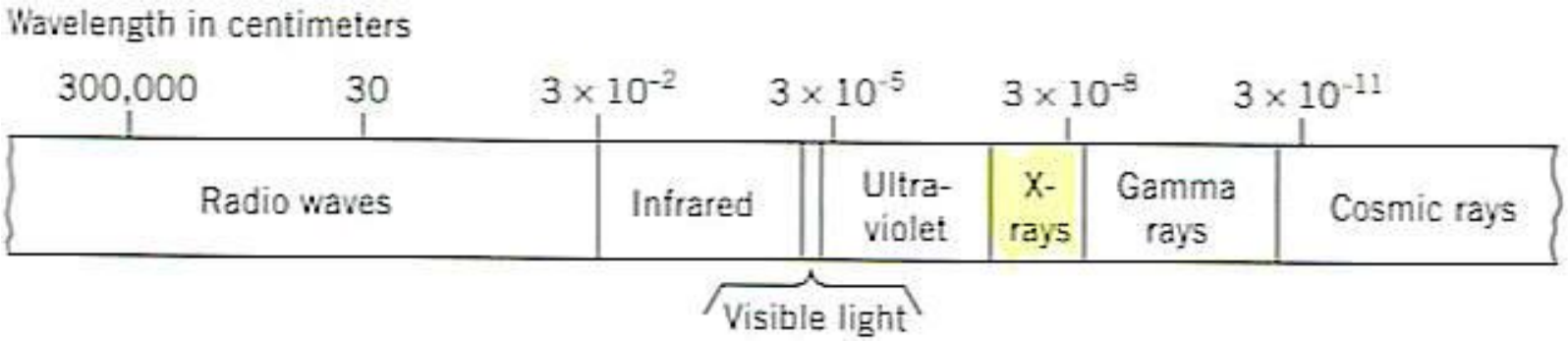


(MED201)



Experiment-2
Determining the crystal structure and lattice parameters of the material from X-ray diffraction patterns

X-Ray Analysis



- X-rays discovered in 1895
- Fundamental to understanding of crystal structure and symmetry
- Powder diffraction analyses are a simple and inexpensive method for identifying minerals, especially fine-grained minerals

X-ray Generation

<u>X-rays</u> – High energy*, highly penetrative electromagnetic radiation

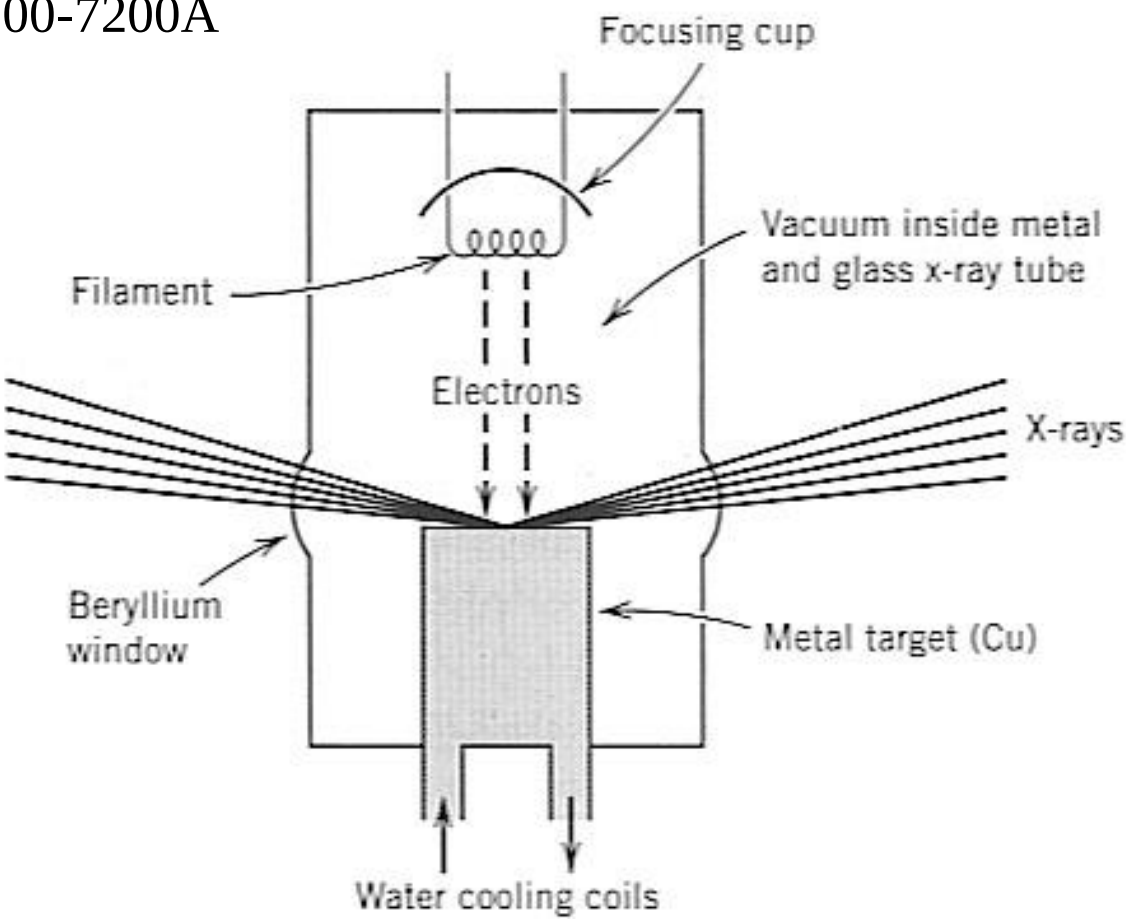
*E = hc/λ

$\lambda_{(X\text{-rays})}$ = 0.02-100Å (avg. ~1 Å)
$\lambda_{(\text{visible light})}$ = 4000-7200Å

<u>X-ray Vacuum Tube</u>

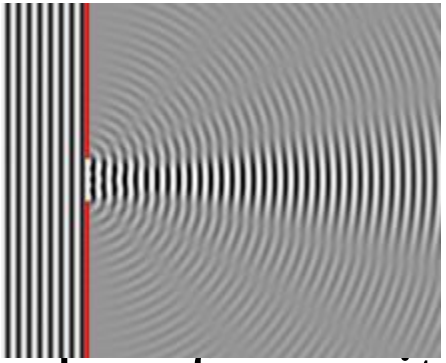
Cathode (W)– electron generator

Anode (Mo, Cu, Fe, Co, Cr) – electron target, X-ray generator



Our instrument uses a copper target

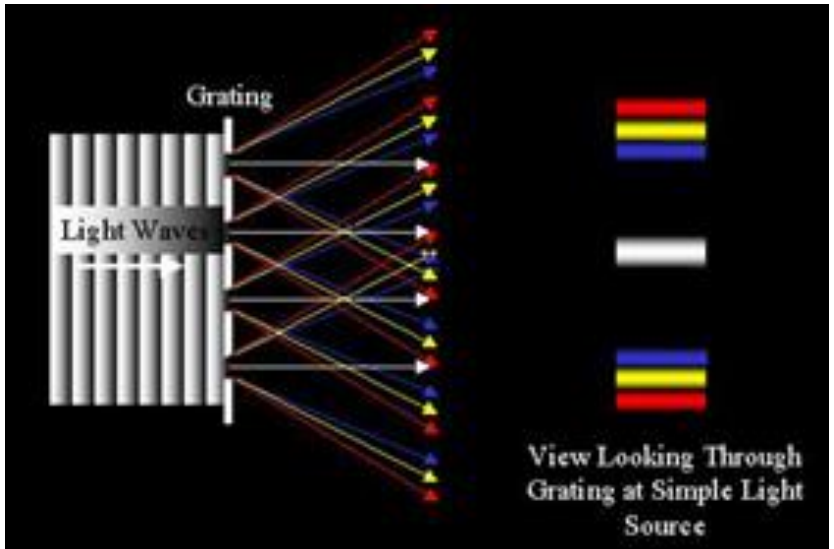
X-ray Crystallography



X-ray wavelengths are on the same order of magnitude as atomic spacings.

Crystals thus make excellent diffraction gratings

Can use the geometry of the x-ray spots to determine geometry of grating (i.e. the crystal)

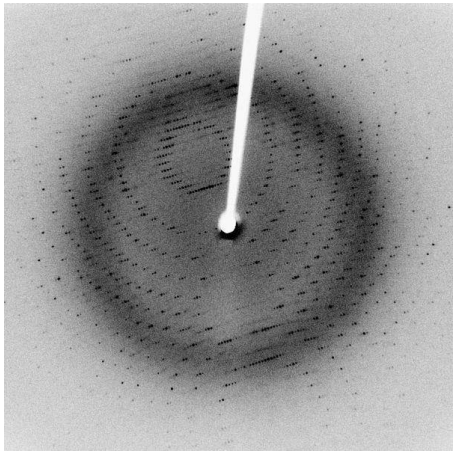


Bragg Diffraction

- Diffraction from a three dimensional periodic structure such as atoms in a crystal is called Bragg Diffraction.
- Similar to diffraction though grating.
- Consequence of interference between waves reflecting from different crystal planes.
- Constructive interference is given by *Bragg's law*:
- Where λ is the wavelength, *d* is the distance between crystal planes, θ is the angle of the diffracted wave. and *n* is an integer known as the *order* of the diffracted beam.

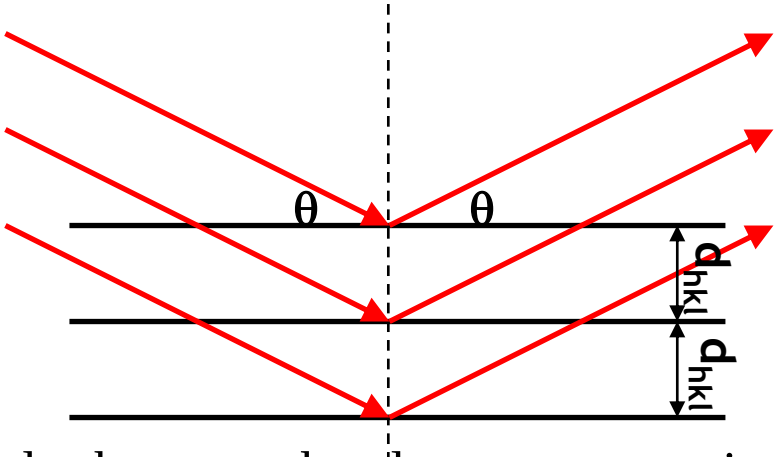
$$n\lambda = 2d \sin \theta$$

Following Bragg's law, each dot (or *reflection*), in this diffraction pattern forms from the constructive interference of X-rays passing through a crystal. The data can be used to determine the crystal's atomic structure.

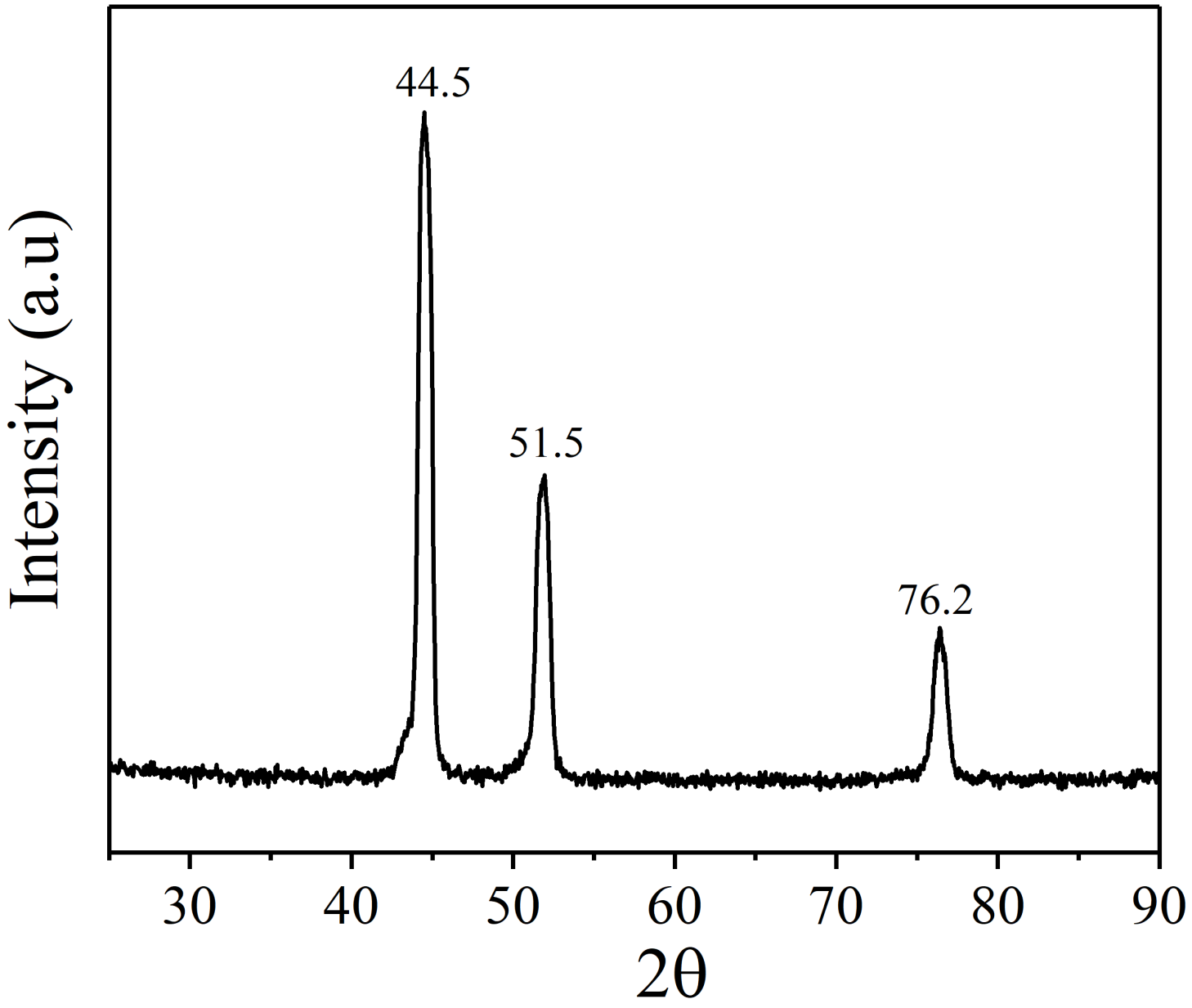


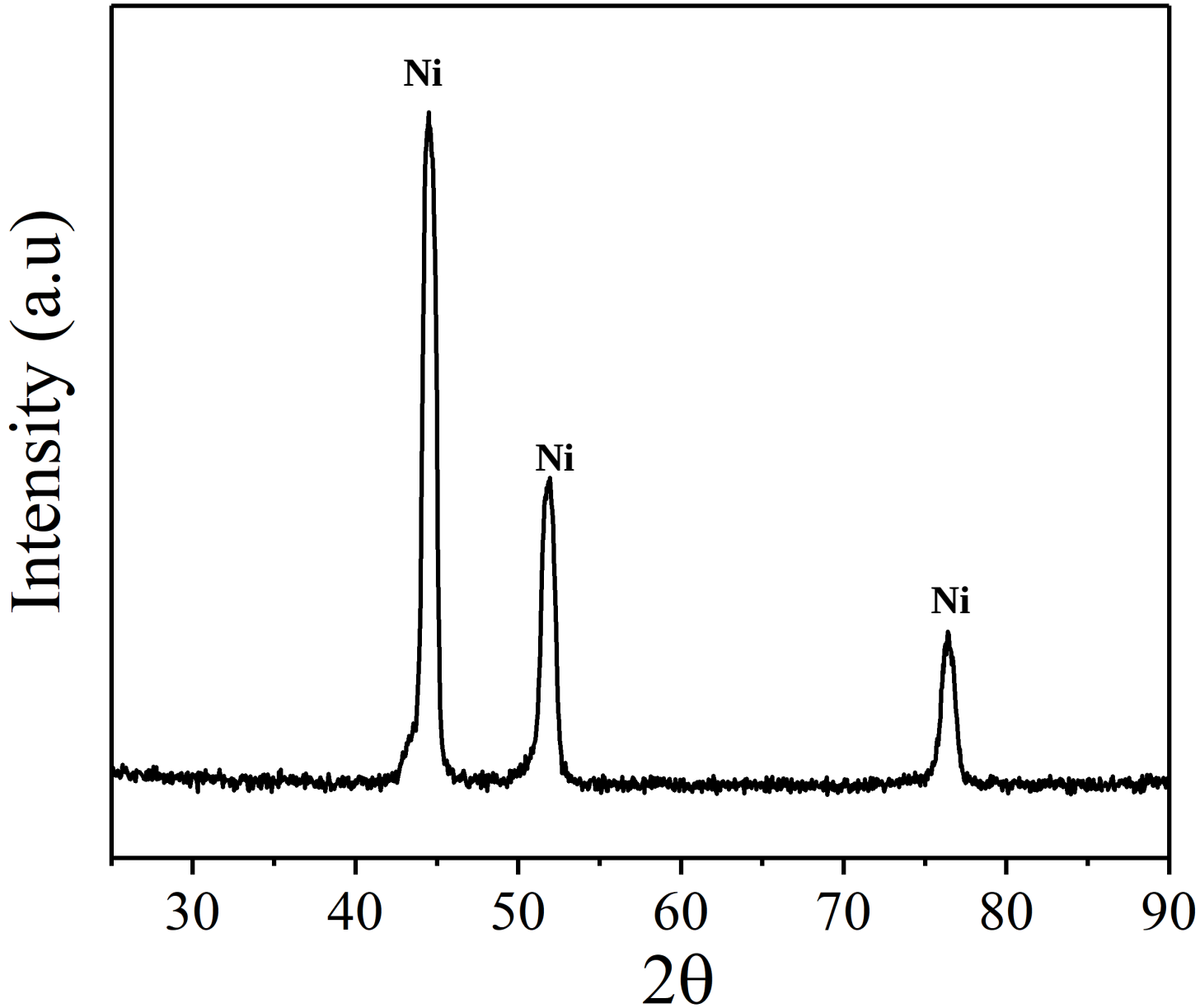
Bragg's law is a simplistic model to understand what conditions are required for diffraction.

$$\lambda = 2d_{hkl}\sin\theta$$



- For parallel planes of atoms, with a space d_{hkl} between the planes, constructive interference only occurs when Bragg's law is satisfied.
  - In our diffractometers, the X-ray wavelength l is fixed.
  - Consequently, a family of planes produces a diffraction peak only at a specific angle
  - Additionally, the plane normal must be parallel to the diffraction vector
    - Plane normal: the direction perpendicular to a plane of atoms
    - Diffraction vector: the vector that bisects the angle between the incident and diffracted beam
- **The space between diffracting planes of atoms determines peak positions.**
- **The peak intensity is determined by what atoms are in the diffracting plane.**


44.5
51.5
76.2
Intensity (a.u)
30
40
50
60
70
80
90
2θ


Ni
Ni
Ni
Intensity (a.u)
30
40
50
60
70
80
90
2θ

Procedure:

- Determine the 2θ values for each peak
- Calculate the value of $\sin^2\theta$ is calculated for each diffraction line
- Use different set of integers given in Table 1 to determine the values for column 4 of the observation table
- Determine the set of integers which results in a constant value for in column 4 of the observation table
- Determine the miller indices corresponding to the value of s from Table 2
- Based on the determined set of the integers from the previous step, determine the crystal structure with the help of Table 1 and 2
- Determine the value of lattice parameter '*a*' in column 5 after substituting the value of wavelength (0.154 nm for Cu-Kα source) for each peak
- Select the value of a corresponding to the highest angle peak

Table 1

Crystal structure	Value of sum $s = (h^2 + k^2 + l^2)$
Simple cubic	1,2,3,4,5,6,8,9,10,11,12,13,14,16, ...
Body-centered cubic	2, 4, 6, 8, 10, 12, 14, 16, . . .
Face-centered cubic	3, 4, 8, 11, 12, 16, . . .

S.No.	$\sin^2\theta$	s	$\frac{\lambda^2}{4a^2}$	a(nm)	hkl
1					
2					
3					
4					
5					
6					

Iron XRD Data

d (A)	2ø
2.07	43.97
1.47	63.204
1.2	79.854
1.03	95.651
0.92	111.9
0.84	130.358

S.No.	$\sin^2\theta$	s	$\frac{\lambda^2}{4a^2}$	a(nm)	hkl
1					
2					
3					
4					
5					
6					

Result

- The crystal structure of Iron is _____
- The lattice parameter of the Iron is _____

Cr XRD Data

d (A)	2ø
2.05	44.17
1.45	64.23
1.18	81.58
1.03	96.91
0.92	113.84
0.84	133.20

S.No.	$\sin^2\theta$	s	$\frac{\lambda^2}{4a^2}$	a(nm)	hkl
1					
2					
3					
4					
5					
6					

Result

- The crystal structure of Cr is _____
- The lattice parameter of the Cr is _____